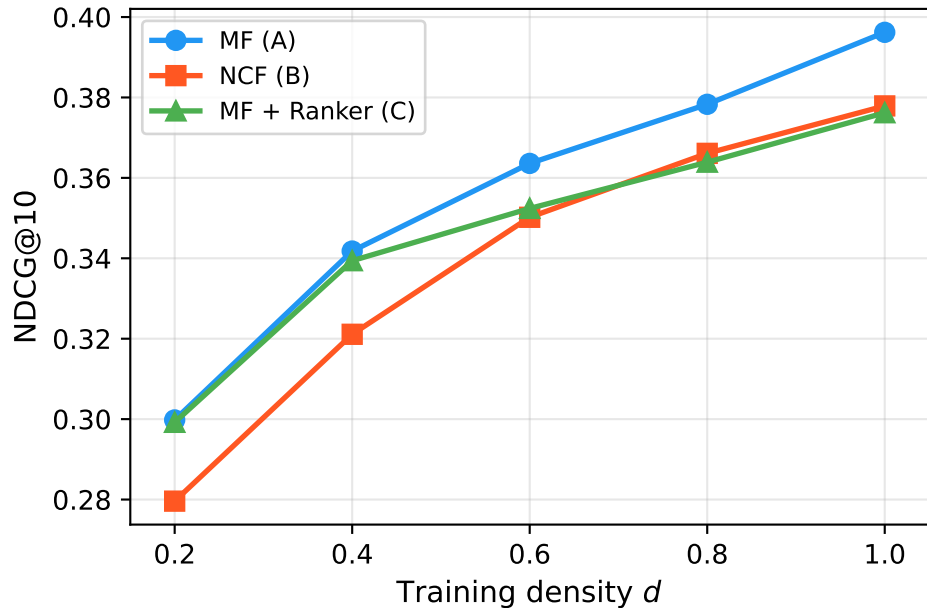


When Does Ranking Help? Sparsity Analysis on MovieLens 1M

NDCG@10 vs. Training Density



HR@10 vs. Training Density

